

11/07/26

Short Answer Question

classmate

Date _____

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1. What is Data Science? Explain its primary objective.

Data Science is an interdisciplinary field that combines statistics, mathematics, computer science, and machine learning to extract meaningful insights from data. It helps organizations analyze both structured and unstructured data. The primary objective of Data Science is to discover hidden patterns, make accurate predictions, and support better decision-making.

2. Why is Data Science important in today's digital world? Mention any four reasons.

Data Science plays a vital role because huge amounts of data are generated everyday. It helps organizations make informed decisions, automate business processes, improve customer satisfaction, and predict future trends. It is widely used in healthcare, finance, education, and e-commerce to solve real-world problems.

3. List any four benefits of Data Science in business and society.

Data science improves business efficiency by analyzing customer behaviour and market trends. It helps detect fraud, optimize operations and increase profits. In society, it supports disease prediction, traffic management and disaster forecasting. It also enables better policy-making through data-driven insights.

4. Mention four real-world applications of Data Science.

Data Science has applications in various fields such as healthcare, finance, education and transportation. It is used for recommendation system in Netflix & Amazon, fraud detection in banking, disease diagnosis in hospitals and weather forecasting. Self-driving cars & virtual assistants also rely on Data Science techniques.

5. What are the different Facets of data? Briefly explain each.

The main facets of data are structured, semi-structured and unstructured data. Structured data is stored in tables with rows & columns. Semi

6. Difference between structured, semi-structured and unstructured data with suitable examples.

Structured data has a fixed format. Such as databases. Semi-structured data has a flexible structure, like XML & JSON files. Unstructured data has no fixed format, such as images, videos, & emails.

7. What is Big Data? State its key characteristics.

Big data refers to very large and complex datasets. Its five characteristics are Volume, Velocity, Variety, Veracity, & Value. These features help in handling & analyzing massive amounts of data.

8. Explain the Big data ecosystem. Name any four technologies used in it.

The big data ecosystem includes tools used to collect, store, process, & analyse large datasets. Common technologies are Hadoop, Apache Spark, Hive, and HBase.

9. What are the major phases of the Data Science process?

The main phases are data collection, data preparation, data analytics, model building, model evaluation, deployment and presentation of results. Each phase helps in extracting meaningful insights from data.

10. What is data retrieval? Mention two common sources from which data can be retrieved.

Data retrieval is the process of obtaining data from storage systems for analysis. Common sources include database, csv files, web APIs, and websites. The retrieved data is then used for processing.

11. Why is data collection considered an important step in the Data Science process?

Data Collection is the first step in the Data Science process. Accurate and relevant data improves the quality of analysis and model performance. Poor-quality data can lead to incorrect results & decisions.

12. What is data cleansing? List any four common data-cleansing techniques.

Data cleansing is the process of correcting or removing inaccurate and incomplete data. Common techniques include removing duplicate records, handling missing values, correcting errors, and standardizing data formats.

13. What is data integration? Why is it necessary in Data Science?

Data integration is the process of combining data from multiple sources into a single dataset. It provides a complete view of the data, removes redundancy, and improves the accuracy of analysis.

14. Explain data integration? Why is it necessary in Data Science?

Data transformation is the process of converting data into a suitable format for analysis. Examples include converting text into numbers, changing date formats, and normalizing data values. It improves data quality and consistency.

15. What is data analysis? Mention any four techniques used for data analysis.

Data analysis is the process of examining data to identify useful information.

patterns. Common techniques include statistical analysis, regression analysis, clustering and data visualization.

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